

1 Documentation conventions

The STM32F405xx/07xx, STM32F415xx/17xx, STM32F42xxx and STM32F43xxx devices have an Arm^{®(a)(b)} Cortex^{®(c)}-M4 with FPU core.



1.1 List of abbreviations for registers

The following abbreviations are used in register descriptions:

read/write (rw)	Software can read and write to these bits.
read-only (r)	Software can only read these bits.
write-only (w)	Software can only write to this bit. Reading the bit returns the reset value.
read/clear (rc_w1)	Software can read as well as clear this bit by writing 1. Writing '0' has no effect on the bit value.
read/clear (rc_w0)	Software can read as well as clear this bit by writing 0. Writing '1' has no effect on the bit value.
read/clear by read (rc_r)	Software can read this bit. Reading this bit automatically clears it to '0'. Writing '0' has no effect on the bit value.
read/set (rs)	Software can read as well as set this bit. Writing '0' has no effect on the bit value.
read-only write trigger (rt_w)	Software can read this bit. Writing '0' or '1' triggers an event but has no effect on the bit value.
toggle (t)	Software can only toggle this bit by writing '1'. Writing '0' has no effect.
Reserved (Res.)	Reserved bit, must be kept at reset value.

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1.2 Glossary

This section gives a brief definition of acronyms and abbreviations used in this document:

- The CPU core integrates two debug ports:
 - JTAG debug port (JTAG-DP) provides a 5-pin standard interface based on the Joint Test Action Group (JTAG) protocol.
 - SWD debug port (SWD-DP) provides a 2-pin (clock and data) interface based on the Serial Wire Debug (SWD) protocol.

For both the JTAG and SWD protocols, please refer to the Cortex[®]-M4 with FPU Technical Reference Manual
- Word: data/instruction of 32-bit length.
- Half word: data/instruction of 16-bit length.
- Byte: data of 8-bit length.
- Double word: data of 64-bit length.
- IAP (in-application programming): IAP is the ability to reprogram the flash memory of a microcontroller while the user program is running.
- ICP (in-circuit programming): ICP is the ability to program the flash memory of a microcontroller using the JTAG protocol, the SWD protocol or the bootloader while the device is mounted on the user application board.
- I-Code: this bus connects the Instruction bus of the CPU core to the Flash instruction interface. Prefetch is performed on this bus.
- D-Code: this bus connects the D-Code bus (literal load and debug access) of the CPU to the Flash data interface.
- Option bytes: product configuration bits stored in the flash memory.
- OBL: option byte loader.
- AHB: advanced high-performance bus.
- CPU: refers to the Cortex[®]-M4 with FPU core.

1.3 Peripheral availability

For peripheral availability and number across all STM32F405xx/07xx and STM32F415xx/17xx sales types, please refer to the STM32F405xx/07xx and STM32F415xx/17xx datasheets.

For peripheral availability and number across all STM32F42xxx and STM32F43xxx sales types, please refer to the STM32F42xxx and STM32F43xxx datasheets.